

Auto Email / Ticket Categorizer

AI/ML Technical Assessment — One-Page Project Summary

1. Objective

Built a lightweight NLP system that reads incoming support-ticket text and automatically routes it to **Billing, Technical, HR, or General**. The design emphasizes fast inference, defensible predictions, and safe handling of uncertain tickets.

2. Dataset

Item	Details
Dataset	Synthetic / dummy labeled support-ticket dataset
Size	998 tickets ; 3 columns: ticket_id, ticket_text, department
Classes	Billing, Technical, HR, General
Quality	No missing values; duplicate records checked

3. Preprocessing & Feature Engineering

Ticket text was normalized using **lowercasing**, URL removal, removal of non-alphabetic characters/numbers, and whitespace normalization. Clean text was transformed with **TF-IDF** using unigram/bigram features and sublinear term frequency. The vectorizer was fitted only on training data to prevent data leakage.

4. Model & Training

Logistic Regression was selected because it is fast and effective for high-dimensional sparse text features and provides class probabilities through `predict_proba()`. A stratified **80/20 train-test split** was used. TF-IDF and the classifier were combined in a Scikit-learn Pipeline. A lightweight **GridSearchCV with 5-fold cross-validation** was used during experimentation to tune TF-IDF and classifier parameters using macro F1.

5. Evaluation & Metrics

Metric / technique	Purpose
Accuracy	Overall percentage of correctly classified tickets.
Precision / Recall / F1	Per-class performance; macro F1 balances performance across departments.
Confusion Matrix	Shows correct classifications and class-to-class errors.
5-fold CV	Used during tuning to assess robustness on training data.

Observed hold-out result: Accuracy = **100%**; macro F1 = **1.000**. Because the synthetic dataset contains distinctive department-specific vocabulary, this is an optimistic result and should not be interpreted as production-level accuracy.

6. Real-Time & Edge-Case Handling

Implemented a real-time prediction function that accepts a new subject/body, applies the same preprocessing and TF-IDF transformation, and returns the predicted category. A **confidence score** is returned with each prediction. If confidence is below **60%**, or the ticket is extremely short, the system sends it to a **Manual Review Queue** rather than auto-assigning it. A keyword-based layer also tags tickets as **URGENT** or **NORMAL**. A CLI demo was added for live inference.

7. Techniques & Outcome

Techniques: text normalization, TF-IDF, n-grams, sparse linear classification, stratified splitting, Pipeline, GridSearchCV, 5-fold CV, macro F1, confusion-matrix analysis, probability-based confidence, threshold-based human escalation, rule-based priority tagging, and real-time CLI inference.

Outcome: Demonstrated an end-to-end lightweight NLP triage workflow from raw ticket text to automated department routing, with a practical human-in-the-loop fallback for uncertain cases.

Technology Stack: Python • Pandas • NumPy • Scikit-learn • Matplotlib